Pau de las Heras Molins

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PUBLICATIONS

2026 F. Roldan Sanchez, **P. de las Heras Molins**, D. Fridovich-Keil, and G. Bakirtzis. "Cross-entropy optimization with prioritized constraints". *Preprint*

B. Lechardoy, **P. de las Heras Molins**, T. Lahire, L. Pautet, D. Filliat, D. Fridovich-Keil, and G. Bakirtzis. "Cascaded consensus splitting for multi-branch contingency games". *Preprint*

P. de las Heras Molins, B. Yalcinkaya, L. Peters, D. Fridovich-Keil, and G. Bakirtzis. "Controllability in preference-conditioned multi-objective reinforcement learning". *International Conference on Neuro-Symbolic Systems (NeuS)*

2025 **P. de las Heras Molins***, E. Roy-Almonacid*, D. H. Lee, L. Peters, D. Fridovich-Keil, and G. Bakirtzis. "Approximate solutions to games of ordered preference". *IEEE International Conference on Intelligent Transportation Systems (ITSC)*

TALKS

2026 Machine learning from role models
École polytechnique (Cosynus Seminar, LIX)

Do preferences control behavior?
Inria Lille (Scool team)

Preference in agent control
UC Berkeley (ICON lab), MIT (Zardini group, LIDS), UT Austin (Center for Autonomy)

Controlling multi-objective reinforcement learning agents through preference
UMich (Inigo Incer's group), Iowa State University (Cody Fleming's group)

EDUCATION

2026– PhD in Computer Science, *Institut Polytechnique de Paris*

2026 Reinforcement Learning Summer School (RLSS), *Politecnico di Milano*

2025 IEEE RAS Summer School on Multi-Robot Systems, *Czech Technical University*

2024–26 MSc in Machine Learning & Cybersecurity, *Universitat Politècnica de Catalunya (UPC)*

2018–19 Erasmus, *Eidgenössische Technische Hochschule Zürich (ETH)*

2015–19 BSc in ICT Systems Engineering, *Universitat Politècnica de Catalunya (UPC)*

MENTORSHIP

2026 Aristide Bailly (Engineering School)
Controllable Self-Play for Highway Merging (coadvised with Georgios Bakirtzis)

Khaled Abou Khachfe (Stage M2)
Reinforcement Learning for Robust Control (coadvised with Georgios Bakirtzis)

EXPERIENCE

Institut Polytechnique de Paris

2026 Research Engineer
Funded by the "Architecture des Systèmes Complexes" chair, part of INFRES at Télécom Paris.

Universitat Politècnica de Catalunya (UPC)

2023–24 Research Associate
Worked on the research project "MobileWuR: Pushing the limits of energy efficiency in mobile devices through wake-up radio solutions".

2021–23 Adjunct Professor
Taught courses on informatics, systems integration, applications and services on the internet, and embedded systems.

UVE Solutions

2024–25 Software Engineer
Develop and maintain backend software for the main data ingestion, storage and exploitation pipeline. Deploy and manage services for external consumption.

2019–24 Technical Consultant Developer
Implemented software for data extraction, transformation, and loading (ETL). My responsibilities also included requirements gathering and customer management.

REFERENCES

Georgios Bakirtzis, Assistant Professor
Télécom Paris, Institut Polytechnique de Paris
bakirtzis@ip-paris.fr
Relationship: MSc thesis and PhD supervisor

David Fridovich-Keil, Assistant Professor
The University of Texas at Austin
dfk@utexas.edu
Relationship: Research collaborator